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MINISTRY OF ENERGY AND NATURAL RESOURCES
TRASHIGANG FOREST DIVISION
TRASHIGANG RANGE OFFICE
UDZORNG BEAT OFFICE



LEMONGRASS MANAGEMENT AND MARKETING PLAN
UDZORONG SORBANG DETSHEN
{1st Revision}



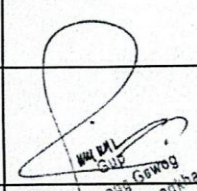
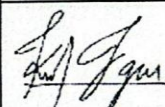
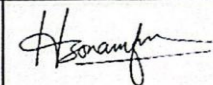
PERIOD: JANUARY 2026 - DECEMBER 2035
UDZORONG GEOG
TRASHIGANG DZONGKHAG

Appendix I: NWFP Management and Marketing Plan Approval Sheet

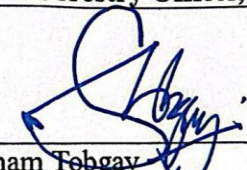
APPROVAL SHEET

Name of NWFP Management and Marketing Group:	Udzorong Sorbang Detshen
Village(s):	Rizor, Barkazor & Lenchi
Gewog:	Udzorong
Dzongkhag /Dungkhag	Trashigang
No. of households:	11HH
Area (Ha)	276.5 Ha
Plan period:	2026-2035

Prepared and submitted for approval

	Chairperson of NWFP Management and Marketing Group	Gup of the Gewog	Head FRPM section of Division/Park	Chief Forestry Officer of Division/Park
Signature:				
Name:	Cheten Norbu	Dorji Tshering	Kelzang Jigme	Sonam Tobgay
Date:	22/12/2025	22/12/2025	22/12/2025	22/12/2025

Reviewed and recommended for approval

	Name of reviewing officer, FRPMD	Chief Forestry Officer, FRPMD
Signature		
Name	Kezang Wangmo	Sonam Tobgay
Date	23/12/2025	24/12/25

Approved by:

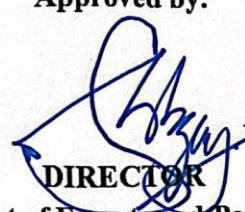

 DIRECTOR 24/12/25
 Department of Forests and Park Services

Table of Contents

PART A: NWFP MANAGEMENT AND MARKETING PLAN.....	5
1. Background	5
2. Goal and objectives:.....	5
3. Description of NWFPs present in the area demarcated for management.	6
4. Participatory mapping and calculation of the area.....	6
5. Resource assessment	7
6. Harvesting prescriptions for sustainable NWFP harvesting	8
7. Marketing and sales plan	8
8. Marketing arrangements	8
9. Benefits to the members.....	9
10. Benefits to the group	10
11. LG collection within the Community Forest	11
12. Collection of firewood for the distillation	11
PART B: GROUP BY-LAWS	12
Members of the NWFP management & marketing group	12
Executive committee.....	12
Incentives to the committee members.....	14
Membership fee and purpose	14
Group Fund management and Utilization.....	14
Conflict resolution	15
Meeting and Purpose.....	15
Penalties	15
Roles and responsibilities of other stakeholders	16
Termination / Cancellation/ Dissolve/ Group Deregistration	16

List of Figures

Figure 1. Map showing the area managed by the Lemongrass Management Group.....	6
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List of Tables

Table 1. Zone distribution for distillation drums and group members	7
Table 2. Overview of the lemon grass production.	7
Table 3. Brief description of NWFPs for commercial harvesting in each season.	8
Table 4. Prescription for sustainable harvesting of NWFPs	8
Table 5. Marketing and sales plan for the group.....	8
Table 6. Financial projection and benefits to the members from NWFP sales	9
Table 7. Financial projection of the group, the community development fund.....	10

PART A: NWFP MANAGEMENT AND MARKETING PLAN

1. Background

The Lemon grass management and marketing plan for Udzorong Sorbang Detshen is a cross-cutting management regimes within the jurisdiction of the Divisional Forest Office, Trashigang. The identified collection and harvesting area for this cross-cutting plan falls inside the Udzorong Community Forests and State Reserved Forest Land. The Community-Based Lemongrass Management Plan for Udzorong Gewog represents a strategic initiative aimed at fostering the sustainable utilization of lemongrass resources through the active involvement of local communities. Lemongrass, which predominantly grows under Chirpine (*Pinus roxburghii*) forest cover, serves as a vital source of income for the community.

A Gewog-wide Lemongrass Management Plan was previously implemented for the entire Udzorong Gewog for the period 2007 to 2017. However, after its expiry, the plan was not revised. Instead, separate lemongrass management groups were formed for Baepam and Gengkhar villages, leaving the rest of Udzorong without any formal management plan.

To address this gap and bring additional areas under sustainable management, this plan proposes the formation of a new lemongrass management group comprising members from Rizor, Barkazor, and Lenchi villages. The goal is to improve coordination and promote sustainable harvesting practices within these areas.

This management plan outlines a structured and participatory approach to resource governance. It emphasizes regulated harvesting, conservation of the resource base, and equitable benefit-sharing among community members. The management group currently consists of 11 members and oversees a total area of 276.5 hectares. Based on resource assessments and yield estimates, the area is projected to produce approximately 1,692 kilograms of lemongrass oil annually, with an estimated market value of Nu. 32,14,800.

This plan serves as a guiding document to ensure that lemongrass harvesting in Udzorong is both economically viable and ecologically sustainable, thereby supporting community livelihoods while conserving natural resources for future generations.

2. Goal and objectives:

1. To sustainably manage the lemon Grass,
2. To generate income through sale of Lemon grass oil.
3. To elevate poverty and improve the livelihood of community
4. To create employment opportunities for the youth.

3. Description of NWFPs present in the area demarcated for management.

Lemongrasses belong to the family of Poaceae (Graminae). The perennial plants grow in clumps of up to two meters in diameter, while the leaves are up to one meter in length. The grass survives for a period of at least six years. Lemongrass is aromatic grass native to Southeast Asia, known for its lemon-scented leaves and stalks. It is widely used in culinary dishes, teas, and herbal remedies due to its refreshing flavor and medicinal properties. Lemongrass is also valued in the cosmetic industry for its essential oil, which is used in aromatherapy, massage oils, and skincare products. Additionally, it has antimicrobial and anti-inflammatory benefits, making it popular in traditional medicine for digestive health, stress relief, and immune support.

4. Participatory mapping and calculation of the area

The lemongrass distillers have collaborated to create a sketch map outlining the customary areas used by each distillation drum. Based on this sketch, ground validation of the area boundaries and subdivision of the lemongrass extraction zones have been carried out using GPS and SW Map.

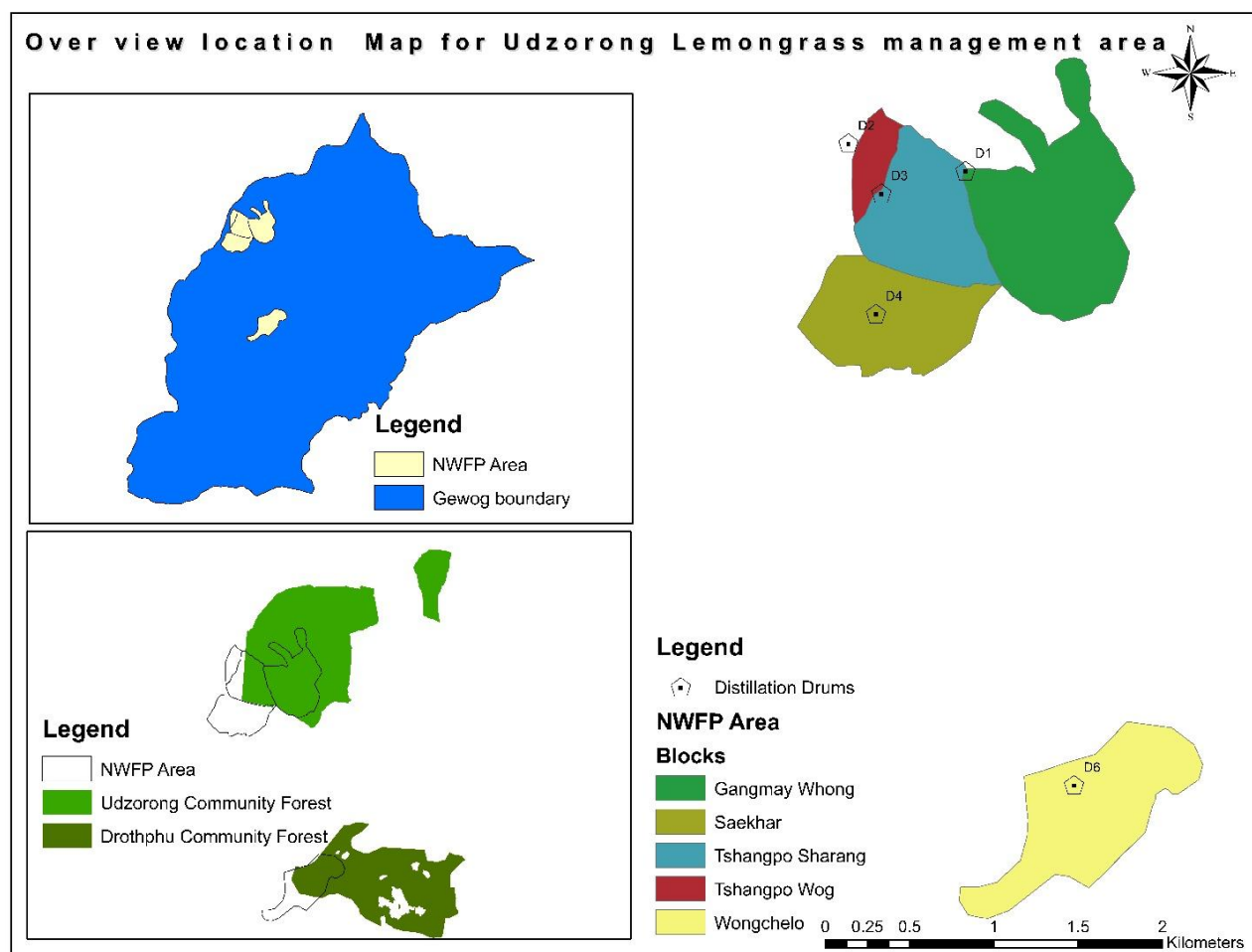


Figure 1. Map showing the area managed by the Lemongrass Management Group.

Table 1. Zone distribution for distillation drums and group members

Zone/Area Name	No. of Drums	Drum Owner	Members
Gangmay Wong	1	Pema Tenzin	
Tshangpo Sharang	1	Cheten Norbu	Tenzin Dorji
Tshangpo Wog	1	Thucken	Tsheten Wangdi
Saekhar	1	Tashi Tenzin	Kenzang Penjore & Tenzin Norbu
Wongchelo	1	Yeshe Dorji	
Wongchelo	1	Tshewang Tobgay	Kezang Namgay

5. Resource assessment

The resource assessment was conducted in accordance with the *Guidelines for Resource Assessment and Management of Lemongrass* (2008). In the absence of proper historical data on lemongrass oil production from the area, the estimation was based on existing research mentioned in Guidelines, which indicates that the oil yield from wild lemongrass stands can reach up to 9 kg per hectare. For planning purposes, it has been assumed that approximately 40% of the total area is operable for lemongrass harvesting, considering unproductive zones such as barren lands, steep slopes, and other inaccessible or unsuitable areas. Second harvest yields are assumed at approximately 70% of the first harvest.

Table 2. Overview of the lemon grass production.

Zone	Area (Ha)	1st cut (Kg)	2nd cut (Kg)	Annual Oil Production (Kg)
Gangmay Whong	91.4	329	230	559
Tshangpo Sharang	48.5	175	122	297
Tshangpo Wog	11.3	41	28	69
Saekhar	55	198	139	337
Wongchelo	70.3	253	177	430
Total	276.5	996	696	1692

Table 3. Brief description of NWFPs for commercial harvesting in each season.

NWFP name	Harvesting area(s) (name)	Parts harvested	Use/purpose (traditional & others)	Harvesting season (months)	Estimated quantity for harvest in a season/year
<i>Cymbopogon flexuosus</i>	Udzorong	Upper part	Essential oil	June-November	1692

6. Harvesting prescriptions for sustainable NWFP harvesting

Table 4. Prescription for sustainable harvesting of NWFPs

NWFP species	Harvesting prescriptions for sustainable harvesting
<i>Cymbopogon flexuosus</i>	• Collect lemongrass at maximum 2 times per year • Cut the grass at a height of 20 cm. • Leave at least 3 to 4 flower stalks/tillers per entire clump.

7. Marketing and sales plan

Currently, LG Oil is supplied exclusively to Bio-Bhutan. The group aims to obtain certification under the Organic Standard of Bhutan and to standardize packaging, with the goal of expanding the market across the country and establishing a presence in international export markets.

Table 5. Marketing and sales plan for the group

NWFP	Potential market/buyer	Estimated quantity for sale per season (unit)	Form of Products (dry/fresh)	Expected price per unit	Total (Nu)
Lemon grass	Bio-Bhutan	1692kg	Oil	1900	32,14,800

8. Marketing arrangements

The Lemon Grass will be collected by the group once the collection date is announced by the Chairman, and the extracted oil will be taken to each member's house. The Forest Office will announce the date for inspection, during which the products will be examined, and Certificates of Origin will be issued to facilitate royalty payments according to the prevailing rates, in the name of the group. Each member will oversee the weighing of their products to ensure accuracy.

Afterwards, members can market their products individually or collectively, with the members' consent, upon obtaining an Internal Movement Order (NWFP).

9. Benefits to the members

There are several benefits an individual member can gain, few are provided below:

- ✓ Improved and stable income for each member sale of lemon grass
- ✓ Right to ownership and access to resources (lemon grass)
- ✓ Enhance self-capacity through capacity building and skill development
- ✓ Reduce the cost of expenditure through sharing resources (transportation, processing facilities, storage facilities, etc.)
- ✓ Stronger access to market and negotiation power with supply chain actors
- ✓ Access to financial and institutional support
- ✓ Risk sharing and livelihood resilience
- ✓ Sustainable management of resources
- ✓ Gain self-voice in decision making

Table 6. Financial projection and benefits to the members from NWFP sales

Sl. No	Particulars	Products (names)	Total Amount (Nu)
		Lemongrass oil	
1	Revenues		
a	Total quantity (unit) -	1692kg	-
b	Market price (Nu per unit) -	1900	-
c	Total revenues from forest product sales (a*b)		32,14,800
A	Total revenues		32,14,800
2	Expenses		
a	Packaging materials	-	-
b	Royalty	1692 * 5	8460.00
c	Transportation costs		4,000.00
d	Compensation fee to the executive committee for marketing products		
e	Contribution to community development fund		32,148(1% of total sale)
f	Incidental expenses		5,000.00
g	Membership fee	500/member	5,500.00
B	Total expenses		42,648.00

C	Total benefit (net income) of group members (Total revenues A-Total expenses B)		31,72,152.00
D	Average income to group members (C/no of group members)		2,88,377.45
E	Price per unit paid to group members (C/1a		1874.79

10. Benefits to the group

The NWFP management group can have more benefits compared to individual members:

- ✓ Stronger recognition from relevant organizations that can strengthen the responsibility, benefit sharing and transparency of the group initiatives
- ✓ As a collective effort, the group can sustainably plan and harvest the resources, monitor and control overharvesting ensuring the long-term availability of resources and conservation of the habitat
- ✓ The group can collectively aggregate the resources and reduce the cost and expenditure that can be incurred during processing, packaging, storage and transportation
- ✓ The branding and marketing of the products can be improved and expanded through quality assurance, certification and competitiveness in both domestic and international market
- ✓ The group is more likely to receive support and investments from donors, government agencies, NGOs and financial institutions
- ✓ The group can also formally collaborate with the government agencies, private sector, buyers and service providers thereby receiving support in technology transfer and product development
- ✓ The group can collectively manage risk and plan better risk management strategies
- ✓ Knowledge sharing and innovation for the groups through peer learning, adoption of improved management practices and innovations
- ✓ Long-term organizational sustainability through clear procedure, benefit sharing mechanisms and income generating activities

Table 7. Financial projection of the group, the community development fund

SL no.	Particulars	Amount (Nu)
	Revenues	
1	Membership registration fees	5500.00
2	Contribution to the Community Development Fund from sales of product	32,148
A	Total Revenue	37,648.00
	Expenses (only include expenses that will be paid from the group funds)	

	TADA to the Committee and other members for their travel on behalf of the group.	5000.00
B	Total Expenses	5000.00
C	Balance: Total revenues -Total expenses	32648.00

11. LG collection within the Community Forest

Lemongrass collection within the Community Forest (CF) will be permitted only to individuals who are CF members of the respective location. Each member must obtain a collection permit by paying a lump-sum compensation of Nu. 1,500 to the CF before collecting the resources.

12. Collection of firewood for the distillation

Given the remote locations of the distillation drums, procuring firewood from commercial suppliers is not feasible. Therefore, firewood must be sourced from nearby State Reserved Forests. For distillation, drums situated within CF areas, firewood shall be obtained from the CF, subject to the payment of applicable royalty to the Community Forest management group.

13. Monitoring and Evaluation

Monitoring will be conducted annually, with frequency adjusted, if necessary, based on the committee's decision. The primary objective is to assess resource management, detect threats such as diseases, and prevent illegal activities within the management area. After each monitoring, findings will be presented to group members to identify solutions and actions. Any illegal activity must be reported immediately to the Udzorong Beat Office for necessary action. Joint monitoring will be carried out by the group and the Beat and Range office to evaluate the implementation of the management plan, by-law compliance, financial performance, record-keeping, and overall group governance. This includes physical verification of harvesting blocks to ensure adherence to management prescriptions. The Beat Office will prepare an annual monitoring report for submission to the Range and Divisional Forest Office, Trashigang, and the Department of Forests and Park Services (DoFPS). Additionally, a midterm review will be conducted in the fifth year and a final evaluation in the ninth year by the Forest Monitoring and Information Division (FMID), DoFPS.

14. Waste Management

Effective waste management within NWFP management areas is crucial to ensure environmental sustainability and resource protection. Special emphasis will be placed on managing non-degradable waste generated from various activities. In accordance with the Forest and Nature Conservation Rules and Regulations 2023 and the Waste Prevention and Management Regulation 2012, all NWFP group members are responsible for maintaining cleanliness and ensuring that waste is properly handled. Any waste produced during harvesting, processing, or related operations must be collected, segregated, and disposed of appropriately following the provisions of the Waste Prevention and Management Regulation 2012. Regular awareness and monitoring will also be carried out to promote responsible waste management practices among members.

PART B: GROUP BY-LAWS

Name of the Group	Udzorong Sorbang Detshen		
Members of the NWFP management & marketing group	Village	No. of member households	Geog
	Rizor, Barkazor, Lenchi	11 HH	Udzorong
	Responsibilities and rights of members: ❖ Managing and harvesting NWFPs in a sustainable manner. ❖ Cross-check illegal activities within the members and by intruders. ❖ Each member has right to harvest the Lemongrass from the designated area allocated to them within the timeline framed in the bylaws. ❖ Each member has the right to know about the status of the financial transparency matter/CDF utilization. ❖ Membership has the right to re-elect executive members depending upon the membership will and urgency.		
Executive committee			
Chairperson	Name of Chairperson: Cheten Norbu Contact No.: 17432148		
	Responsibilities of the chairperson: ❖ Coordinate and preside over general meetings and draw conclusions of the discussions based on majority consensus ❖ Plan and formulate annual work plans in consultation with other management committees ❖ Resolve conflict of the members wherever possible and forward to higher authority as per the rules and procedures prescribed under the section Conflict Resolution of these By-Laws. ❖ Exploring market in consultation with the relevant Agency or organization ❖ Responsible for endorsing cheques for withdrawing group funds along with the secretary and treasurer ❖ Make necessary arrangements for the visitor to visit the group ❖ Enforcing fines and penalties for illegal activities ❖ Review and monitor the planned activities ❖ Shoulder the responsibilities of Secretary and Treasurer in their absence ❖ Submit annual progress to relevant organizations / agencies.		
Secretary	Name of the Secretary: Thukten Contact No.: 17993452		

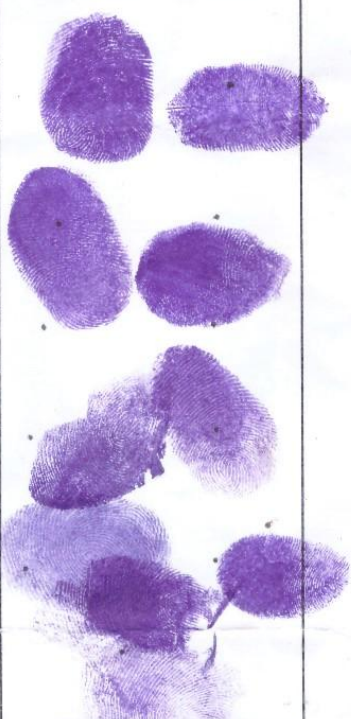
	Responsibilities of the Secretary: ❖ Officiate the Chairperson in his absence ❖ Responsible for all correspondence related to the group ❖ Keep all necessary records with regards to implementation of group activities ❖ minutes of all the meetings ❖ Co-sign cheques for withdrawing group fund along with the Chairman and Treasurer ❖ Responsible for preparing annual work plan and progress reports for submission to the relevant organization / Agency or the Department ❖ Resolve conflict of the cooperative members ❖ Enforcement of fines and penalties for illegal activities ❖ Keep inventory of all tools and equipment ❖ Shoulder the responsibilities of Treasurer in his absence
Treasurer	Name of the treasurer: Tshewang Tobgay Contact No.: 17993668
	Responsibilities of Treasurer: ❖ Keep all the financial related records and present them to the members during the general meetings ❖ Responsible for withdrawing and depositing of cash in the group Account ❖ Co-sign cheques with the chairperson and the secretary ❖ Officiate as Chairman and Secretary in his/her absence ❖ Resolve conflicts among the members ❖ Enforcement of fines and penalties for illegal activities
Tenure of committee member	❖ The committee members elected from the group members, by the group members unanimously, will serve for 3 years to provide equal opportunity to all members. ❖ However, no executive committee members performing their duties as expected by group members will be terminated before the expiry of the tenure, and the vacant post shall be filled immediately.

Incentives to the committee members	❖ The Travel Allowance (existing taxi local fare) and Daily Allowance (Nu.215) to the Committee and other members for the purpose of group shall be paid from the group fund. However, in case of meetings / workshop / study tour if the travel allowance and daily allowance are paid by concerned organization they will not be entitled from the group fund
Membership fee and purpose	❖ Each member shall contribute a minimum of Nu. 500/- (five hundred) only as a membership fee. ❖ The members who fail to pay the membership fee will be terminated from the group without any benefits. ❖ The new members who are interested & wish to join the group must make payment as per the individual membership fee collected till the date of joining, as per the record mentioned. ❖ Any member (committee / group member) found misusing the group fund will be liable to a penalty of Nu.10% of the total amount misused.
Group Fund management and Utilization	Financial arrangement: The group will have a joint Bank account that can be operated by two people (Chairman and Treasurer). All Bank transactions will have to be well documented, and the financial position (expenditure, income and current balance) must be informed to the groups during meetings. Withdrawals as well as cash deposits must be well maintained in the books of accounts. Fund Management: The sources of funds for the group shall be from: 1. Fines from penalties and absentees 2. Donations from visitors 3. 1% of income to group funds for all the commercial sales of the NWFP All expenses required for smooth functioning of the Group must be authorized by the Executive committee. The Executive Committee submits the financial requirements for the planned activities to the General Assembly. The Bank account must be operated jointly by two members of the Executive Committee. At the closure of the financial year, the accounts are verified by an external person (e.g. extension agent) who will give a report to the General Assembly. Utilization of group funds: ❖ To purchase necessary tools as required by the group

	❖ The fund will be used to meet the administrative and other management costs of the group. ❖ Will be used for serving refreshment for the visitors and guest ❖ Will be used for serving lunch and refreshment during general meetings and for carrying out management activities whenever necessary.
Conflict resolution	Any conflict that arises among the members shall be dealt with by the committees. If the case is beyond the level of the committees, then the case shall be forwarded to the Geog Administration and finally to the district court.
Meeting and Purpose	The General Assembly meets at least twice a year. The Chairperson calls for the General Assembly at least 7 days in advance, furnishing the points of the agenda as well as the time and place of the meeting. The ordinary agenda of the General Assembly is: Presentation and approval of the agenda ❖ Reading the previous minutes and information on the actions taken ❖ Election of the management committee and the other as when necessary ❖ Amend and/or modify the By-laws if required by society ❖ Approval of the terms of reference for the various committees ❖ Presentation and approval of the yearly report and the yearly accounts ❖ Presentation and approval of the business plan and the budget ❖ Information and decision on investments ❖ Miscellaneous decisions which are not covered in the group by-law ❖ Discussion regarding next meeting and tentative date ❖ Every member has one vote in the General Assembly. Decisions of the General Assembly require a two-thirds majority of the members present and voting. ❖ The group will conduct general meetings 2 times a year.
Penalties	❖ Nu.600/- (Six hundred) will be charged as an absentee fee during the general assembly unless the victim undergoes unavoidable circumstances. ❖ The member shall be terminated without any benefit if found committing offences more than 3 times not complying with the group's NWFP management plan and By-laws. ❖ If any of the members resign from the group due to personal issues or unavoidable circumstances, he/she shall submit a

	letter of resignation to the executive members addressing the Chairperson one month before disseminating the reason for the decision. The group must conduct a general assembly and relieve the member with a share of benefits that he/she is liable for till date. The members resigning without having any critical problems shall be subjected to fine of Nu.2000/- (Two thousand) and shall be relieved without any benefits/share. ❖ A member can deploy only 3 people to harvest the grass, failure to abide shall be fine Nu.600/- per person
Roles and responsibilities of other stakeholders	Divisional Forest Office ❖ Technical support ❖ Capacity development ❖ Help in marketing and selling products ❖ Monitoring and evaluation of the group Geog Administration ❖ Help in marketing and selling products ❖ Conflict resolution ❖ Monitoring and evaluation of the group
Termination / Cancellation/ Dissolve/ Group Deregistration	In the event if the group members plan to dissolve, terminate or de-register the group before the expiry of the plan, the following should be abided by the members and respective field offices: ❖ The ownership / NWFP group certificate shall be surrendered to the Division Office ❖ No harvesting or collection of resources shall be permitted after cancellation or deregistration of the group ❖ If the community wishes to collect the resources, the collection of resources should be facilitated strictly by Rule 260 (1&2) ❖ The members shall not be allowed to form a new group or join any NWFP groups ❖ In the event if the NWFP group is dissolved, then 10% of the proceed acquired through sale of NWFPs shall be forfeited to the Government Revenue.

Udzorong Sorbang Detshen Members

Sl. No.	Name	CID no	Gung no.	Contact no.	Signature
1	Cheten Norbu	11515003833	491	17432148	
2	Thucken	11515003662	504	17993452	
3	Tshewang Tobgay	11515003420	385	17993668	
4	Pema Tenzin	11515004064	474	17326285	
5	Tsheten Wangdi	11515003790	483	17452587	
6	Kinzang Penjor	11515003498	434	17426478	
7	Tashi Norbu	11515003830	490	17630637	
8	Tenzin Dorji	11515004097	478	17495576	
9	Tashi Tenzin	11515003650	482	17245253	
10	Kezang Namgay	11515002192	345	17817175	
11	Yeshi Dorji	11515003092	313	17883249	